PO12Q - Quantitative Political Analysis: Uncovering Relationships

Mock Exam 2026/27

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Paper Code: PO12Q –
Exam Period: Mock Exam

Exam Rubric

Time Allowed: 84 Minutes
Exam Type: Written - Standard
Approved Calculators: Permitted

Answer all questions in the answer booklet provided

UNIVERSITY OF WARWICK

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Instructions

- Part I contains the questions for the exam
  - These are divided into four blocks: A, B, C, and D.
  - Block A is compulsory and must be answered by all candidates.
  - You must then select ONE block out of B, C, and D.
  - Should you change your mind during the exam, indicate clearly which block you wish to have marked.
  - You can achieve a maximum of 120 points.
  - Information on the maximum number of points obtainable is outlined at the beginning of each block
- Part II contains the case study to which all questions in Block A of Part I refer
- Part III contains the relevant statistical tables
  - Normal curve tail probabilities are provided in Table 5, page 14.
  - t-distribution critical values are provided in Table 6, page 15.
- Part IV contains the formula collection and notation of the module
  - If you use notation other than in the formula collection (page 17) or notation table (page 21), give an explanation.
- Allowed time: 2 hours.
- Round all intermediary and final results to three decimals.
- You may use a non-programmable, non-graphical scientific calculator. It must not be able to store text or programs, solve equations, or communicate with any other device. This rules out all graphical calculators, including those used at GCSE and A-level. Suitable examples include the Sharp EL-501X, Casio fx-83GTX and Casio fx-85GTX. Other models that meet these requirements are also acceptable. The Casio fx-570 and fx-991 series are not permitted, in any version (including PLUS, X, CW and + models).

For all questions involving calculations

You must show your full working for every calculation. Marks are awarded mainly for working, so an answer given without working will receive little or no credit, even if it is correct. A result read directly from a calculator's statistics mode does not count as working.

For each calculation, working means:

- the formula you are using, written out before you substitute numbers into it;
- the values you substitute into the formula;
- any intermediate results, for example a table of values, deviations from the mean, squared deviations and their sums.

Part I
Exam Questions

1 | Block A

60 points, points for each question in parentheses

All questions in this section relate to the case study in Part II.

Level 1 (each question 4 points)

1. Subsections 13.1 and 13.2: which model building strategy has been employed here? Explain your answer.
2. Subsections 13.1 and 13.2: State the null- and alternative hypotheses underpinning Model 2.

Level 2 (each question 6 points)

3. Interpret the output of Section 11.
4. Subsection 12.1: Can we conclude that the proportion of democracies differs between low- and high-GDP countries in sub-Saharan Africa? Show your reasoning with reference to the output.
5. Table 4: Interpret the intercept in Model 1.
6. Table 4: Interpret the coefficient of "Primary Sector, value added (% of GDP)" in Model 5.
7. Table 4: Interpret the model fit for Model 5.
8. We have applied a logarithmic transformation to per capita GDP. Would a similar transformation make sense for life expectancy? Why / why not?

Level 3 (each question 8 points)

9. Interpret the output in Subsection 14.1. Which next steps do you need to take to address the issue if applicable?
10. Given the output in Subsection **??**: Is the effect size in Model 1 in Table 4 large enough for conventional levels of statistical power? Explain your reasoning and what this means substantively.

2 | Block B

24 points, 6 points for each statement

State whether each of the following statements is true or false. Explain why the statement is true or false.

1. The method of Ordinary Least Squares (OLS) minimises the sum of the distances between the observations and the regression line.
2. It is more sensible to run one multiple regression model with z independent variables than z bivariate regression models.
3. R^2 and Adjusted R^2 ($\bar{R}^2$) cannot be compared.
4. If the values of a variable are highly skewed, this variable cannot serve as an independent variable in an OLS regression.

3 | Block C

24 points, points for each question in parentheses

Table 1 shows the size of the lower chamber of the legislature and the population size of 5 European countries.

1. Specify the null and alternative hypothesis for the relationship between the size of the lower chamber of the legislature and the population size of a country. **(4)**
2. Fit a linear regression model of the type $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ to the data in Table 1. Specify the full sample regression function. **(20)**

Country	Size of legislature	Population in millions
Lithuania	141	2.89
Poland	460	36.5
Austria	183	9.2
Ireland	160	5.42
Germany	736	83.58

Table 1: Regression Data

4 | Block D

24 points, 8 points for each method / statement

Explain each of the following methods or concepts in your own words, and illustrate your explanation with a worked example of your choice.

1. Variance Inflation Factor
2. Homoscedasticity
3. Robust standard errors

Part II
Case Study

5 | Preface

The data set is composed of three different sources joined together:

- World Development Indicators: The World Development Indicators is a compilation of relevant, high-quality, and internationally comparable statistics about global development and the fight against poverty. The database contains 1,400 time series indicators for 217 economies and more than 40 country groups, with data for many indicators going back more than 50 years. (The World Bank, n.d.)
- BMR: This data set provides a dichotomous coding of democracy for every country in the world (222 total) from 1800 to 2020. (Miller et al., 2022)

The data in the data set is for the year 2000.

6 | Code Book

variable	label
`country`	Country Name
`democracy`	0 = Autocracy, 1 = Democracy (Miller et al., 2022)
`gdppc`	GDP per capita (current US Dollars)
`enrol_net`	School enrollment, primary (% net)
`life`	Life expectancy at birth, total (years)
`agri`	Agriculture, forestry, and fishing, value added (% of GDP)

Table 2: Variables in the `ssa` data set for 2000

7 | Preliminaries

```
wdi <- read.csv("ssa.csv")
```

8 | Descriptives for ssa Data Frame

Table 3: Descriptive Statistics for 'ssa' Data Frame

	N	Mean	SD	Min	P25	Median	P75	Max
democracy	46	0.30	0.47	0.00	0.00	0.00	1.00	1.00
gdppc	46	977.44	1447.93	134.54	300.03	465.10	730.35	8063.66
enrol_net	30	64.01	18.96	26.83	53.04	63.63	82.17	99.05
life	46	53.80	6.51	44.83	48.88	52.68	57.44	72.78
agri	40	23.12	15.20	2.61	12.80	22.45	29.97	76.07

9 | Proportion of missing values per variable

```
##   country democracy     gdppc enrol_net      life      agri
## 0.0000000 0.0000000 0.0000000 0.3478261 0.0000000 0.1304348
```

10 | Recoding

10.1 per capita GDP

```
ssa <- ssa %>%
  mutate(gdpcat=
           ordered(
             cut(gdppc, breaks=c(0,350,8064),
                 labels=c("low","high"))))

ssa$log.gdppc <- log(ssa$gdppc)
```

10.2 Life expectancy at Birth

```
ssa <- ssa %>%
  mutate(lifecat=
           ordered(
             cut(life, breaks=c(0,mean(ssa$life),80),
                 labels=c("low","high"))))
```

10.3 Democracy

```
ssa$democracy <- factor(ssa$democracy, labels=c("Autocracy", "Democracy"))
```

11 | Crosstabulations

```
table1 <- with(ssa, table(lifecat, gdpcat))
table1
##         gdpcat
## lifecat low high
##    low   13   14
##    high   3   16

Xsq1 <- chisq.test(table1, correct=FALSE)
Xsq1
##
##  Pearson's Chi-squared test
##
## data:  table1
## X-squared = 5.1477, df = 1, p-value = 0.02328
```

12 | Two-Sample Tests

12.1 Two-Sample Proportion Test

```
table2 <- with(ssa, table(democracy, gdpcat))
table2
##             gdpcat
## democracy   low high
##   Autocracy   9   23
##   Democracy   7    7

prop.test(table2, correct = FALSE)
##
##  2-sample test for equality of proportions without continuity correction
##
## data:  table2
## X-squared = 2.0544, df = 1, p-value = 0.1518
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.52348678  0.08598678
## sample estimates:
##  prop 1  prop 2
## 0.28125 0.50000
```

13 | Regression

13.1 Regression Models

```
model1 <- lm(log.gdppc ~ life, data=ssa)

model2 <- lm(log.gdppc ~ enrol_net, data=ssa)

model3 <- lm(log.gdppc ~ agri, data=ssa)

model4 <- lm(log.gdppc ~ life + enrol_net, data=ssa)

model5 <- lm(log.gdppc ~ life + agri, data=ssa)

model6 <- lm(log.gdppc ~ enrol_net + agri, data=ssa)

model7 <- lm(log.gdppc ~ life + enrol_net + agri, data=ssa)
```

13.2 Results Table

Table 4: Regression Models

	Dependent Variable: log(per capita GDP)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Life expectancy at birth	0.089***			0.060**	0.064***		0.049**
	(0.017)			(0.020)	(0.015)		(0.015)
% Primary School Enrolment (net)		0.034***		0.025**		0.019*	0.010
		(0.007)		(0.007)		(0.008)	(0.008)
Primary Sector, value added			-0.047***		-0.037***	-0.049***	-0.047***
			(0.007)		(0.006)	(0.013)	(0.011)
(Intercept)	1.526	4.229***	7.418***	1.545	3.743***	6.180***	4.015***
	(0.936)	(0.499)	(0.201)	(1.017)	(0.867)	(0.765)	(0.949)
Num.Obs.	46	30	40	30	40	28	28
R2	0.378	0.429	0.524	0.567	0.683	0.702	0.789
R2 Adj.	0.363	0.408	0.512	0.535	0.666	0.678	0.763

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

13.3 Regression Graphs

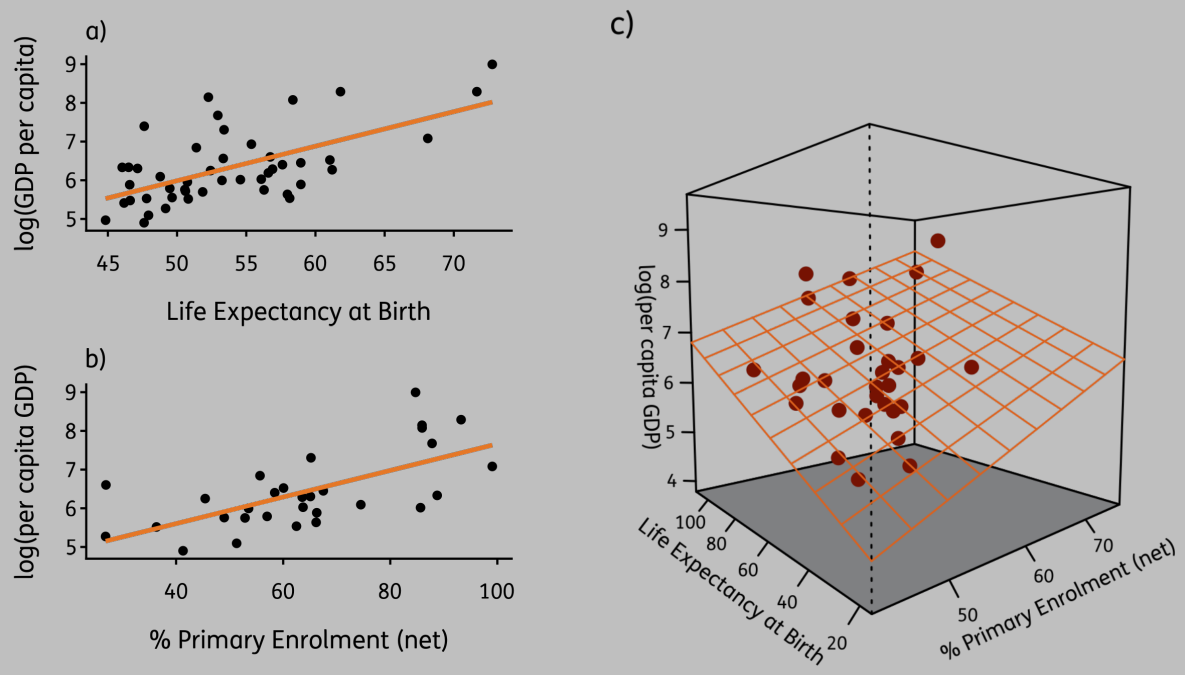


Figure 1: Visualisation of Regression Models

14 | Testing the Classical Linear Assumptions

14.1 `bptest()`

```
bptest(model1)
##
##  studentized Breusch-Pagan test
##
## data:  model1
## BP = 0.29415, df = 1, p-value = 0.5876
```

Part III
Statistical Tables

15 | Normal-Distribution

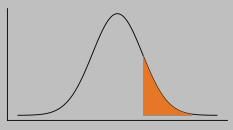


Table 5: Right-Tail Probabilities under the Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002

Table 6: Probabilities under the t-Distribution

Confidence level	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
One-tail probability	$t_{0.25}$	$t_{0.2}$	$t_{0.15}$	$t_{0.1}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$	$t_{0.0005}$
df										
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

17 | χ^2 - Distribution

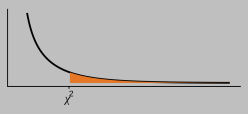


Table 7: Probabilities under the χ^2-Distribution

Upper-tail probability	$\chi^2_{0.995}$	$\chi^2_{0.99}$	$\chi^2_{0.975}$	$\chi^2_{0.95}$	$\chi^2_{0.9}$	$\chi^2_{0.1}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.01}$	$\chi^2_{0.005}$
df										
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.60
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34	12.84
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28	14.86
5	0.412	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09	16.75
6	0.676	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81	18.55
7	0.989	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48	20.28
8	1.344	1.646	2.180	2.733	3.490	13.36	15.51	17.53	20.09	21.95
9	1.735	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67	23.59
10	2.156	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21	25.19
11	2.603	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72	26.76
12	3.074	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22	28.30
13	3.565	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69	29.82
14	4.075	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14	31.32
15	4.601	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58	32.80
16	5.142	5.812	6.908	7.962	9.312	23.54	26.30	28.85	32.00	34.27
17	5.697	6.408	7.564	8.672	10.09	24.77	27.59	30.19	33.41	35.72
18	6.265	7.015	8.231	9.390	10.86	25.99	28.87	31.53	34.81	37.16
19	6.844	7.633	8.907	10.12	11.65	27.20	30.14	32.85	36.19	38.58
20	7.434	8.260	9.591	10.85	12.44	28.41	31.41	34.17	37.57	40.00
21	8.034	8.897	10.28	11.59	13.24	29.62	32.67	35.48	38.93	41.40
22	8.643	9.542	10.98	12.34	14.04	30.81	33.92	36.78	40.29	42.80
23	9.260	10.20	11.69	13.09	14.85	32.01	35.17	38.08	41.64	44.18
24	9.886	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	34.38	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	36.74	40.11	43.19	46.96	49.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89	53.67

Part IV
Formulae and Notation

18 | Formulae

Table 8: Formulae for PO12Q

Statistic	Formula
Correlation Coefficients	
Covariance	$\text{cov}(x, y) = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$
Pearson's r	$r_p = \frac{\text{cov}(x, y)}{s_x s_y}$ $= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
Point-Biserial r_{pb}	$r_{pb} = \frac{\bar{x}_1 - \bar{x}_0}{s}\sqrt{\frac{n_1 n_0}{n(n-1)}}$
Spearman's r_s	$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$
ϕ Coefficient	$\phi = \frac{AD - BC}{\sqrt{(A+B)(C+D)(A+C)(B+D)}}$

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Statistic	Formula
Correlation – Significance Testing	
Noncentrality Parameter (power)	$\lambda = \frac{\rho\sqrt{n-2}}{\sqrt{1-\rho^2}}$
Pearson (Significance)	$t_{\text{obs}} = \frac{r_p\sqrt{n-2}}{\sqrt{1-r_p^2}}$
Point-Biserial (Significance)	$t = \frac{r_{pb}\sqrt{n-2}}{\sqrt{1-r_{pb}^2}}$
Spearman (Significance)	$t = \frac{r_s\sqrt{n-2}}{\sqrt{1-r_s^2}}$
	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$
ϕ (Significance)	$\chi^2 = n\phi^2 \quad (df = 1)$
Regression	
Adjusted R^2	$\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2/(n-k-1)}{\sum(y_i-\bar{y})^2/(n-1)}$
Coefficient of Determination	$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum(y_i-\bar{y})^2}$
Estimated Standard Error of $\hat{\beta}_0$	$\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n\sum(x_i-\bar{x})^2}}\,\hat{\sigma}$
Estimated Standard Error of $\hat{\beta}_1$	$\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum(x_i-\bar{x})^2}}$
	$\hat{\text{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\boldsymbol{X'X})^{-1}$
Estimated Variance	$\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n-p}$
Estimator of β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$
Estimator of β_1	$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^N (x_i-\bar{x})^2}$
	$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$
Explained Sum of Squares	$\sum(\hat{y}_i-\bar{y})^2$
Residual Sum of Squares	$\sum \hat{\epsilon}_i^2 = \sum(y_i-\hat{y}_i)^2$
Total Sum of Squares	$\sum(y_i-\bar{y})^2$

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Statistic	Formula
Variance Inflation Factor	$\text{VIF}_j = \frac{1}{1 - \bar{R}_j^2}$

18.1 Matrices

$$\boldsymbol{X'X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\boldsymbol{X'X})^{-1} = \frac{1}{n\sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\boldsymbol{X'y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Table 9: Notation for PO12Q

Symbol	Explanation
a	A scalar; a single numerical value representing magnitude, written in lower-case italics.
$\boldsymbol{a}$	A vector; an ordered list of numbers (often a column) representing magnitude and direction, written in lower-case bold.
$\boldsymbol{A}$	A matrix; a rectangular array of numbers arranged in rows and columns, written in upper-case bold.
A^{-1}	Inverse of matrix A; the matrix satisfying $AA^{-1} = I$, the matrix analogue of dividing by A.
A'	Transpose of matrix A; the matrix obtained by interchanging its rows and columns.
α	Significance level; the probability of rejecting a true null hypothesis (a Type I error). Common levels are 0.05, 0.01, or 0.001; it also fixes the confidence level $1 - \alpha$.
α_i	Coefficients of an auxiliary (secondary) regression – one regressing an independent variable on the others, used for example to compute the variance inflation factor (VIF) for multicollinearity rather than to answer the main research question.
β	Population regression coefficient; the number multiplied by a variable in the regression equation. A slope coefficient gives the change in y for a one-unit increase in that predictor, holding the others constant.
$\hat{\beta}$	Estimated regression coefficient; the Sample Regression Function estimate of β, obtained by Ordinary Least Squares (OLS), which minimises the sum of squared residuals.
χ^2	Chi-squared test statistic; assesses whether the observed frequencies in a contingency table differ significantly from those expected if there were no association between the variables, $\chi^2 = \sum(O - E)^2 / E$.
d_i^2	Difference in ranks for each pair of observations for Spearman Correlation.
d_{obs}	Observed, standardised effect size (Cohen's d); the magnitude of the difference, expressed in standard-deviation units.
d_{test}	Assumed effect size, for example in a power or sample-size calculation; the value that enters the noncentrality parameter, $\lambda = d_{\text{test}} \cdot \sqrt{n}$.
df	Degrees of freedom, express constraints on our estimation process
E[x]	Expected value of x; the long-run average of a random variable over infinitely many repetitions, i.e. the mean of its probability distribution.
$E[y_i \mid x_i]$	Conditional expectation, reading as 'The expected value of y, given x'.
ϵ	Error term; quantifies the distance between each observation and the corresponding point on the (population) regression line, denoted ϵ_i.
$\hat{\epsilon}$	Residual; an estimate of the error term – the difference between an observation y_i and its estimated value $\hat{y}_i$, denoted $\hat{\epsilon}_i$.
F	F-test statistic; compares explained and unexplained variance to test the joint significance of several coefficients at once.
f^2	Cohen's f^2; the standardised effect size for an F-test, the regression counterpart of Cohen's d. For an overall model $f^2 = \frac{R^2}{1-R^2}$, expressing explained relative to unexplained variance; larger f^2 means a larger effect and thus greater power.
$\frac{\partial f(x)}{\partial x}$	The partial derivative of f of x, with respect to x.
H_0	Null Hypothesis, typically asserting that there is no effect or no difference.
H_1	Alternative Hypothesis, proposes that there is an effect or a difference

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Table 9 – continued

Symbol	Explanation
I	Identity matrix; a square matrix with ones on the diagonal and zeros elsewhere, leaving any matrix unchanged under multiplication.
k	Number of slope coefficients (predictors) in the regression model, excluding the intercept.
K	Total number of regression coefficients including the intercept, so that $K = k + 1$.
λ	Noncentrality parameter; how many standard errors the true effect lies from the null value, $\lambda = d_{\text{test}} \cdot \sqrt{n}$; it centres the non-central t-distribution under the alternative hypothesis.
$\lim_{h \to 0}$	The limit as h approaches zero.
log(x)	Logarithm of the variable x; the exponent to which a base b must be raised to yield x (so $x = \log_b n$ when $b^x = n$). In regression the natural logarithm is typically used, for example for skewed variables.
μ	Population mean of the variable of interest; a parameter – a fixed value characterising the population – estimated by the sample mean $\bar{y}$.
μ_0	Null Hypothesis Value
n	Sample size; the number of observations in the sample.
n_i	Number of cases in each group for Point-Biserial Correlation; n = Total.
Ω	Variance–covariance matrix of the regression errors. Under the standard assumptions $\Omega = \sigma^2 I$, where σ^2 is the mean squared error.
p	p-value, indicates the probability of obtaining a result equal to, or even more extreme than the observed value in the direction of the alternative hypothesis, assuming the null hypothesis is true.
$P()$	Probability of a statement
ϕ	Phi coefficient; measures the association between two binary variables, equivalent to Pearson's r applied to two dichotomies.
r	Correlation coefficient; measures the statistical dependence of two variables by pairwise comparison of values, taking a value between −1 and +1 for strength and direction.
R^2	Coefficient of determination; the proportion of variation in the dependent variable explained by the independent variable(s), $R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$.
$\bar{R}^2$	Adjusted R^2; a version of R^2 that adjusts for the number of independent variables in a multiple regression, giving a more unbiased estimate of the proportion of variance explained.
r_p	Pearson's product-moment correlation coefficient; a parametric measure of the linear association between two variables (assumes approximate normality).
r_{pb}	Point-biserial correlation coefficient; measures the association between a continuous variable and a binary (dichotomous) variable.
r_s	Spearman's rank-correlation coefficient; a non-parametric measure of the monotonic association between two variables, computed on ranks.
ρ	The hypothesised population correlation under H_1.
s	Sample standard deviation; the square root of the variance, $s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$.
se	Estimated standard error; the standard deviation of the sampling distribution of $\bar{y}$ estimated from the sample, $se = \frac{s}{\sqrt{n}}$ (replacing the unknown σ with s).
$\hat{se}$	Estimated standard error; estimates the standard error of a sample regression coefficient.
σ	Residual Standard Error; measures the average distance that the observed values fall from the regression line, expressed in the same units as the response variable.

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Table 9 – continued

Symbol	Explanation
σ^2	Error variance; the variance of the regression disturbances ϵ, assumed constant under homoscedasticity, and estimated by the mean squared error of the residuals.
t_{obs}	Observed, standardised test statistic that measures how many standard errors a sample statistic (typically a sample mean) lies from a hypothesised population parameter.
u	Numerator degrees of freedom of the F-distribution; the number of slope coefficients being tested, $u = k$.
v	Denominator degrees of freedom of the F-distribution; the residual degrees of freedom, $v = n - k - 1$ (equivalently $n - K$).
VcV	Variance–covariance matrix; a square matrix holding the variances of the estimated coefficients on its diagonal and the covariances between pairs of coefficients in its off-diagonal elements.
$\bar{x}_i$	Mean of the continuous variable for group i for Point-Biserial Correlation.
$\bar{y}$	Sample mean; the sum of the observations divided by their number, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$.
$\hat{y}$	A(n estimated) point on the sample regression line.
y_i	The i-th observation in the sample, where $i = 1, 2, \ldots, n$.

Part V
References

List of References

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